

Gradient Descent

CSC 487: Deep Learning
Instructor: Jonathan Ventura

Loss functions

Loss function computes a measure of model “fit” to the data.

Squared error loss function:

$$L(\mathbf{x}^i; \mathbf{w}) = \frac{1}{2}(z^i - y^i)^2$$

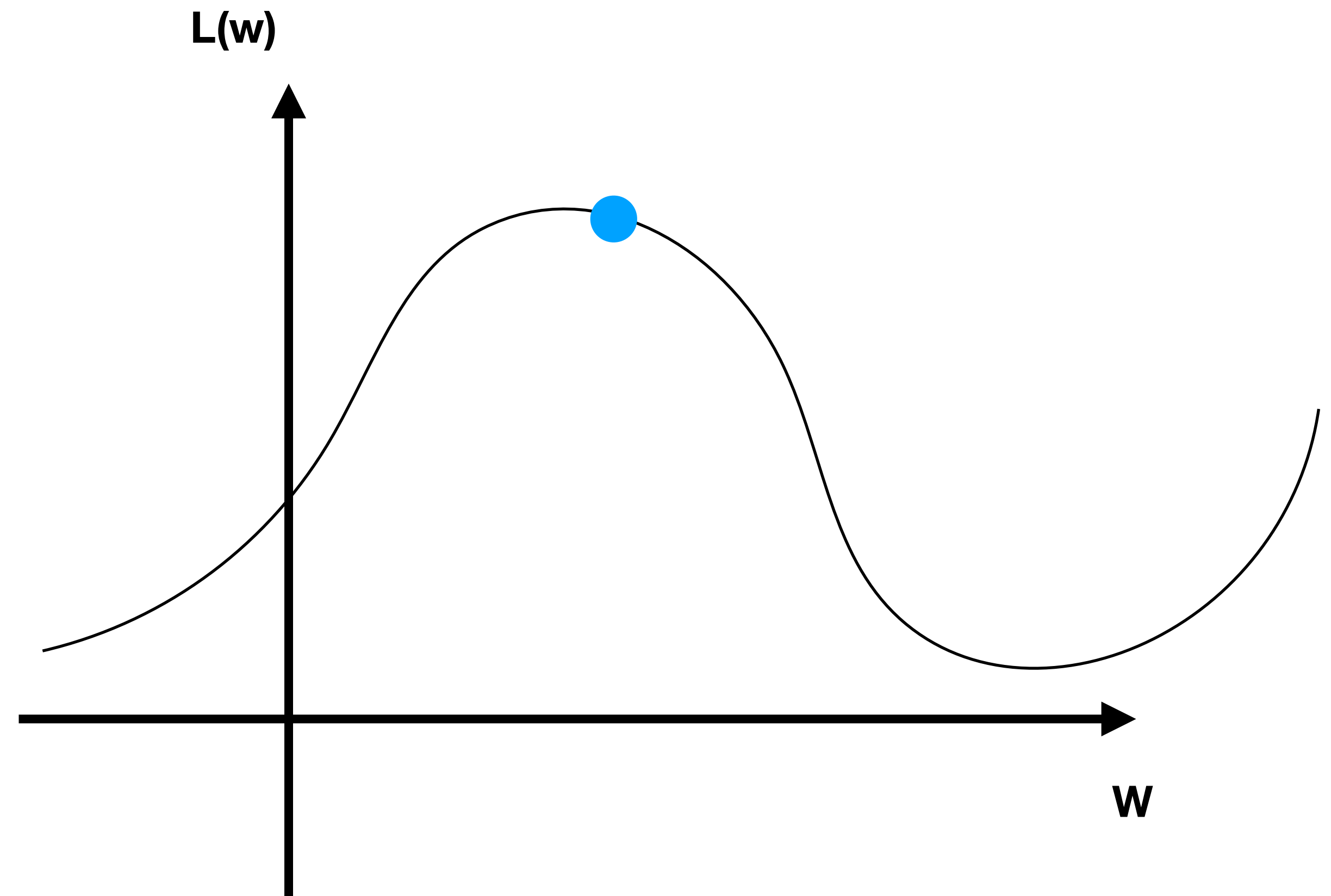
Dot product loss function:

$$L(\mathbf{x}^i; \mathbf{w}) = - (z^i \cdot y^i)$$

We apply **gradient descent** to minimize the loss.

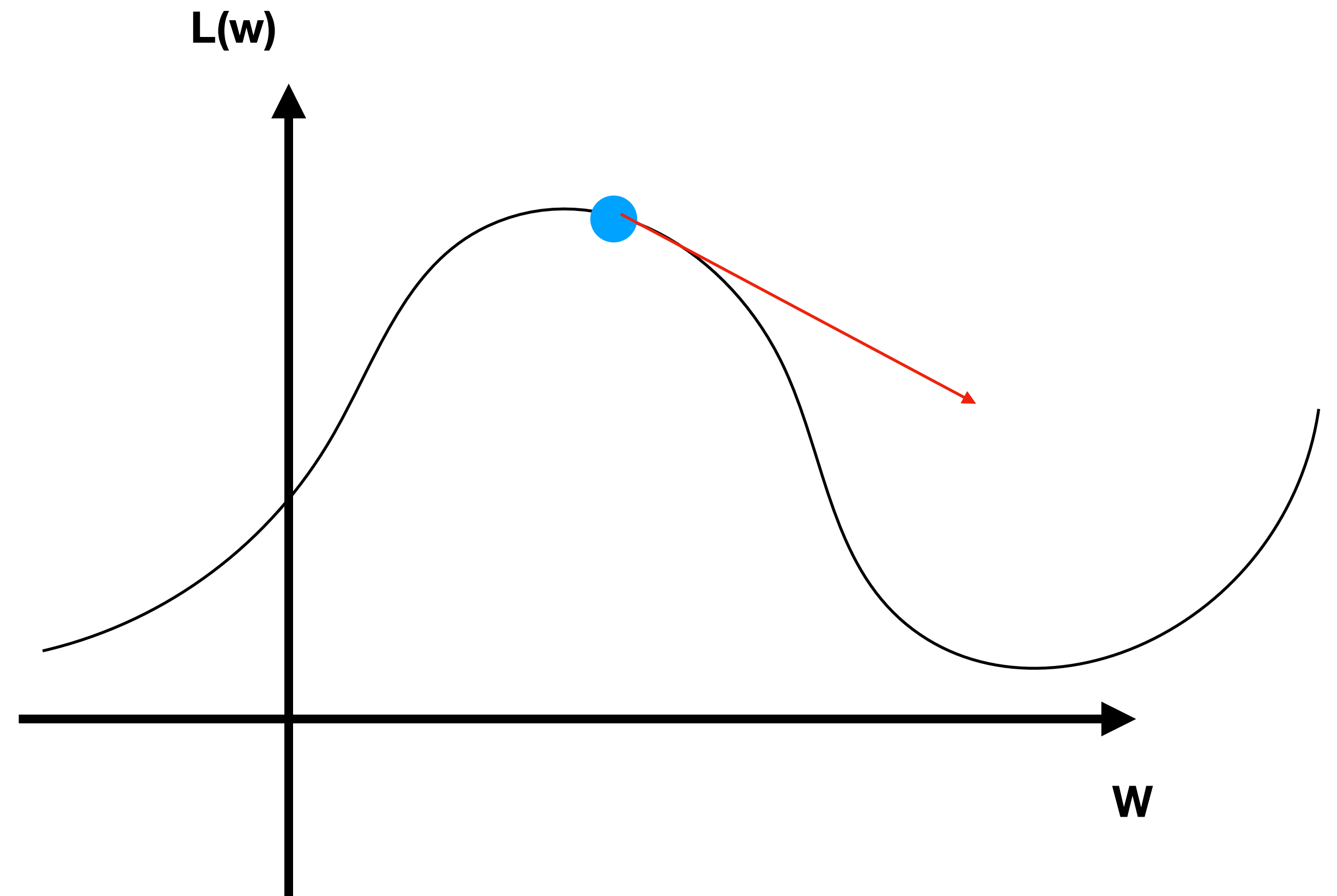
Minimizing the loss function

- Finding a local minimum (1-D example)
- Idea:
 - Start somewhere
 - Take a “step” toward the local minimum
 - Repeat until convergence



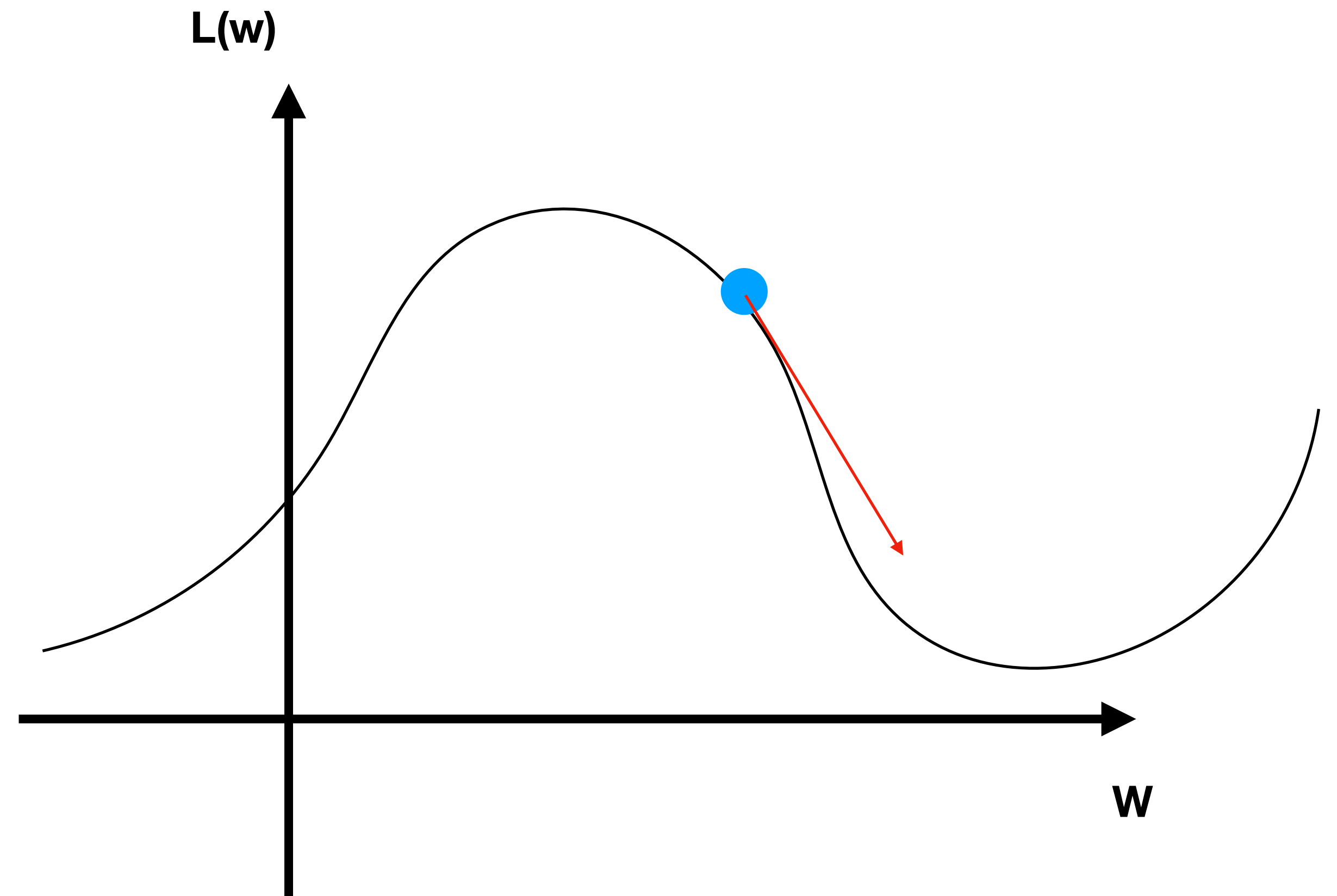
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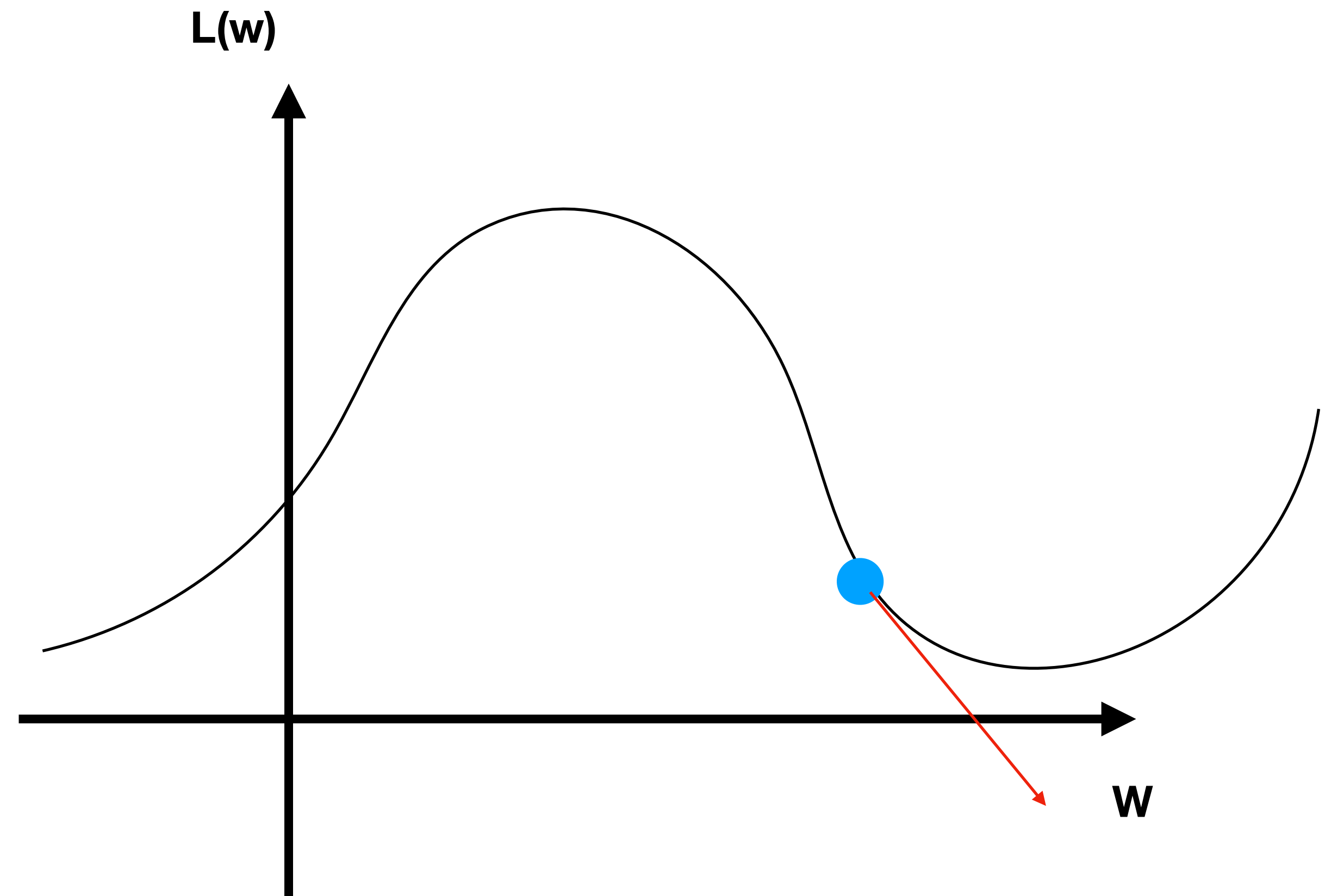
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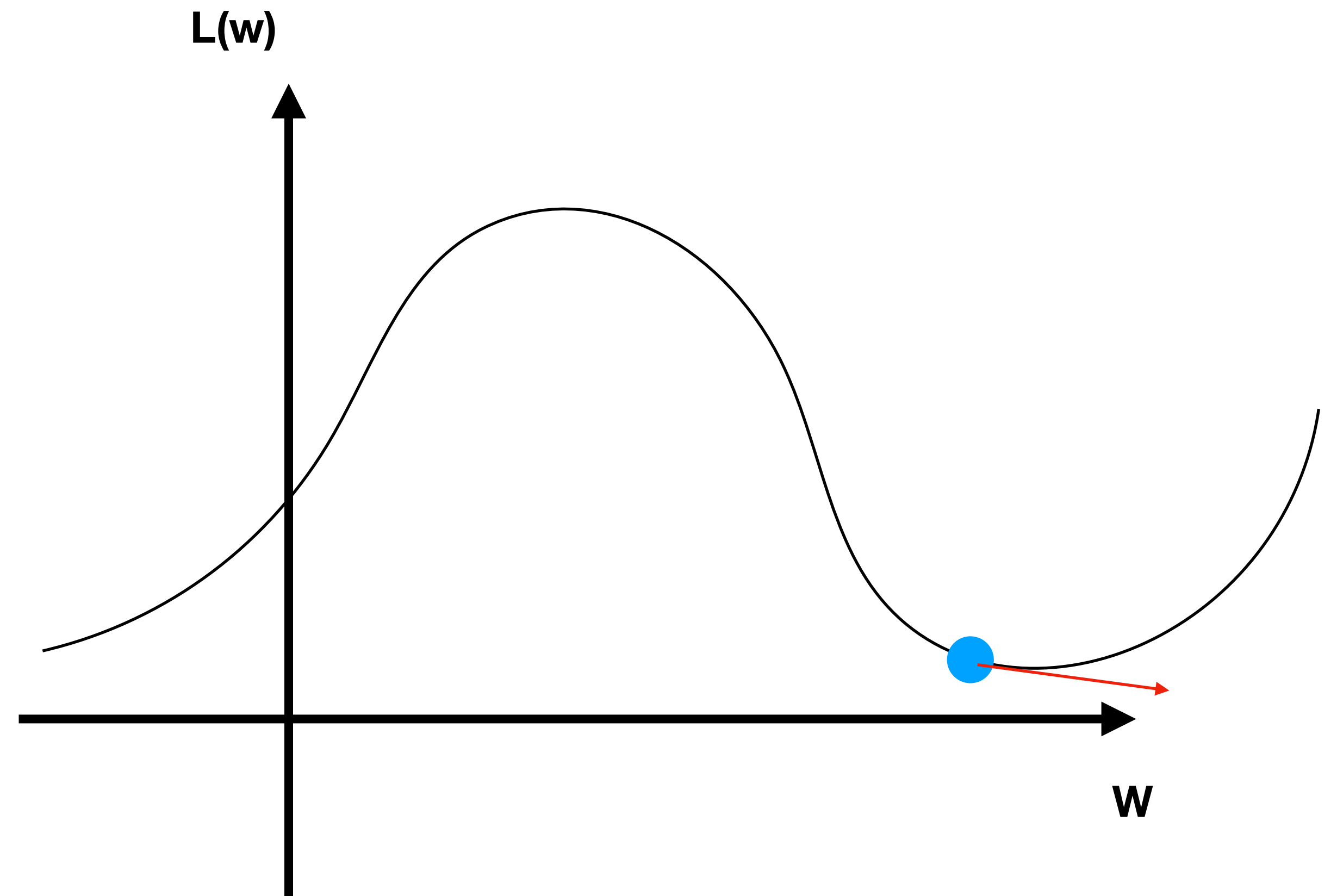
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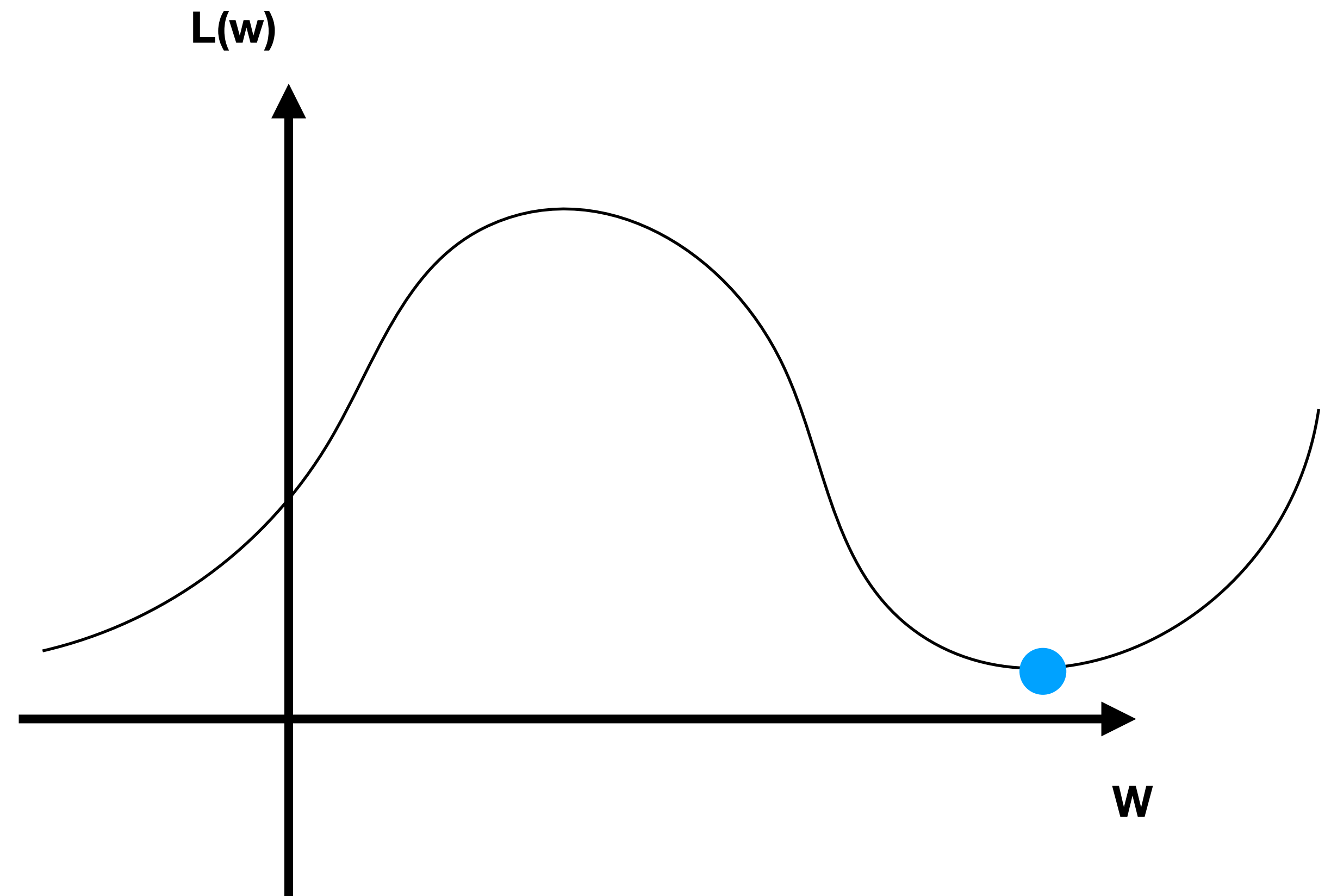
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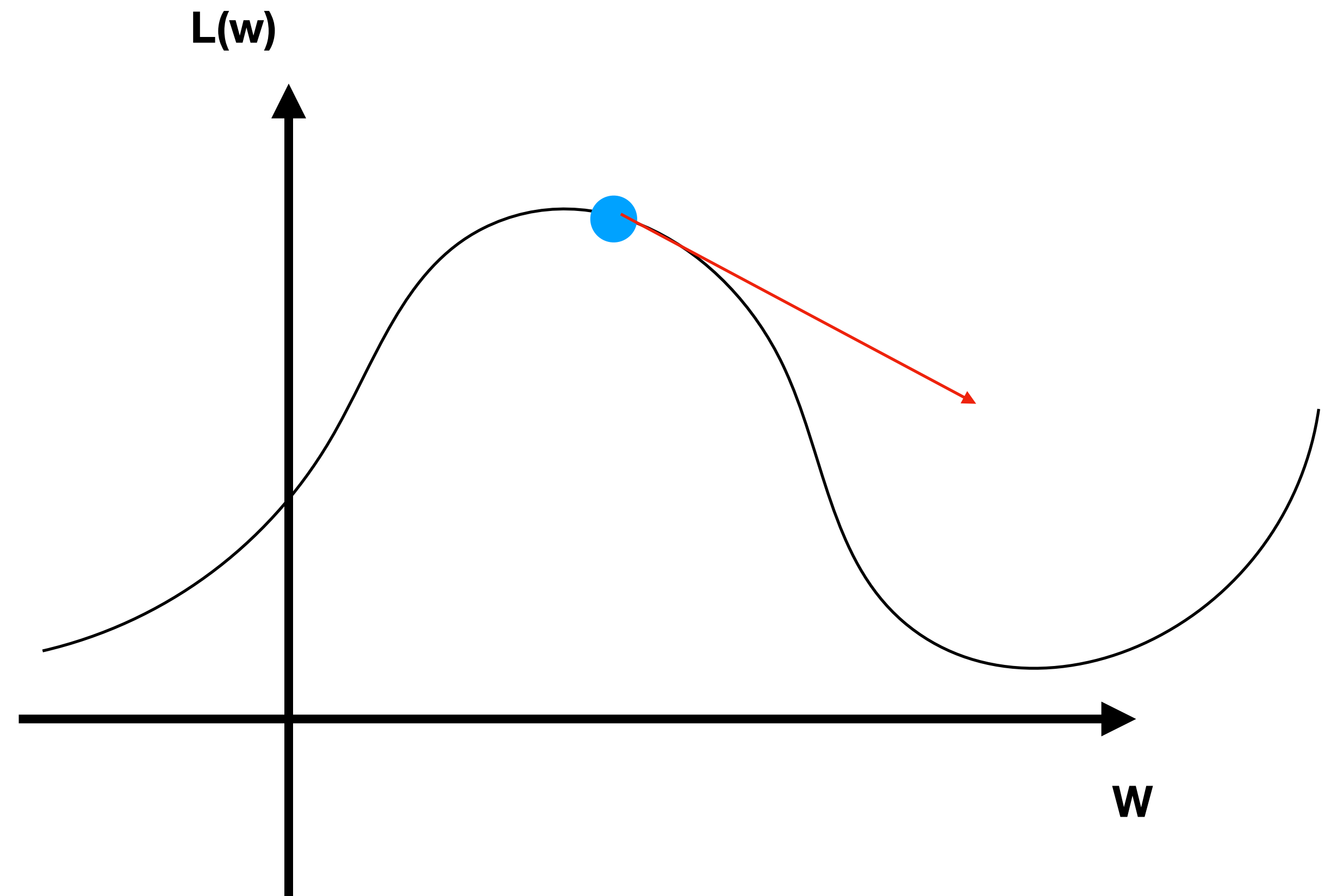


Minimizing the loss function

- Formally:

$$w' = w - s \cdot \frac{dL}{dw}(w)$$

- s is the step size (the “learning rate”)



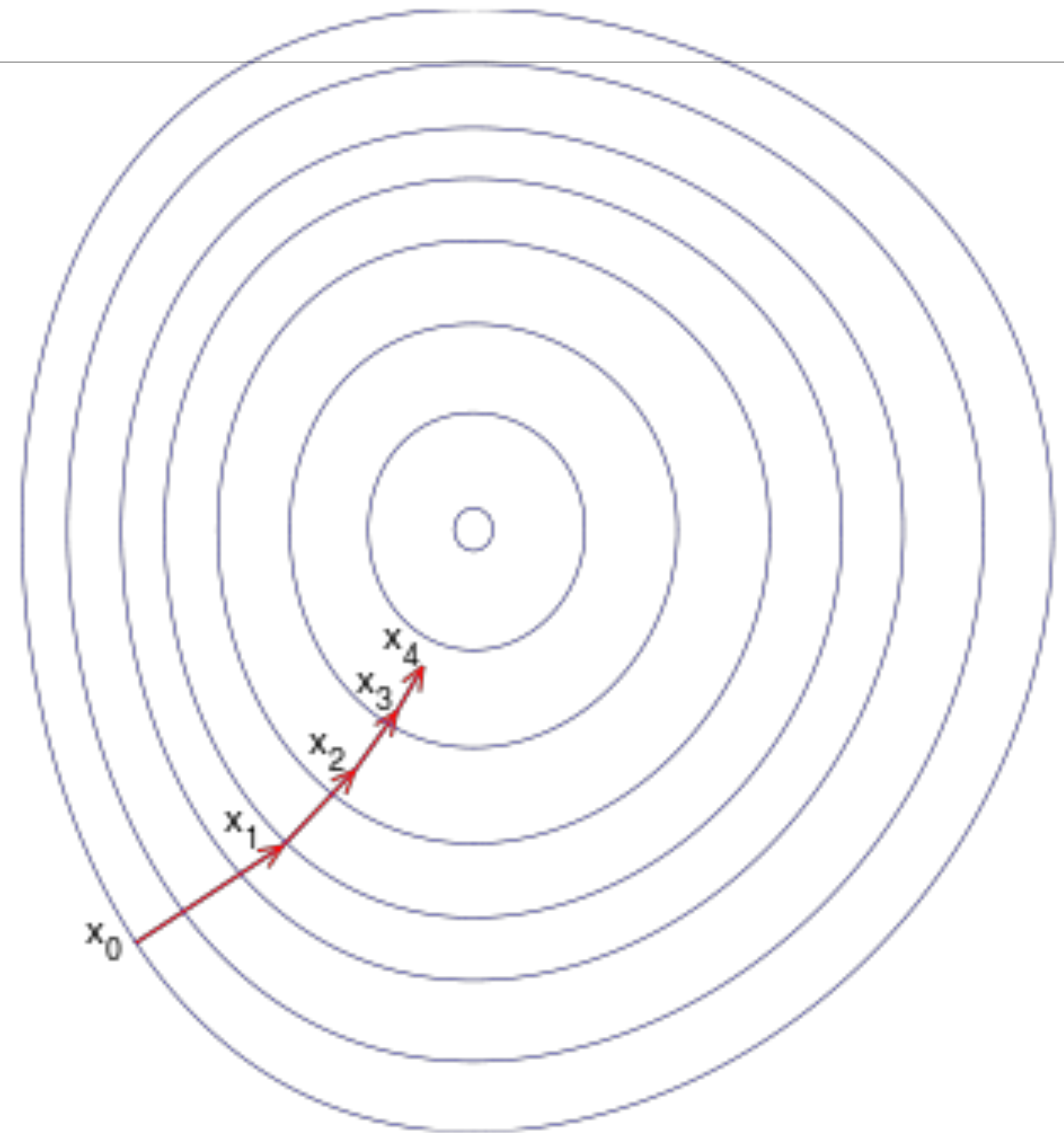
Minimizing the loss function

- With multiple parameters:

$$\mathbf{w} = \mathbf{w} - s \cdot \nabla L(\mathbf{w})$$

- Gradient vector:

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \\ \vdots \\ \frac{\partial L}{\partial w_n} \end{bmatrix}$$



Derivative of loss function

Loss function:

$$L(\mathbf{x}^i; \mathbf{w}) = \frac{1}{2}(z^i - y^i)^2$$

How to compute $\frac{\partial L}{\partial w_j}$?

Chain rule

Recall:

$$\frac{\partial}{\partial x} f(g(x)) = \frac{\partial f}{\partial g}(g(x)) \frac{\partial g}{\partial x}(x)$$

Derivative of loss function

Loss function:

$$L(\mathbf{x}^i; \mathbf{w}) = \frac{1}{2}(z^i - y^i)^2$$

$$L(z) = \frac{1}{2}(z(\mathbf{x}; \mathbf{w}) - y)^2$$

$$z(\mathbf{x}; \mathbf{w}) = \mathbf{w}^T \mathbf{x}$$

Derivative of loss function

Loss function:

$$L(\mathbf{x}^i; \mathbf{w}) = \frac{1}{2}(z^i - y^i)^2$$

Compute $\frac{\partial L}{\partial z}$ and $\frac{\partial z}{\partial w_j}$.

Derivative of loss function

Loss function:

$$L(\mathbf{x}^i; \mathbf{w}) = \frac{1}{2}(z^i - y^i)^2$$

$$\frac{\partial L}{\partial z} = (z - y)$$

Derivative of loss function

Score function:

$$z^i = w_0 x_0^i + \cdots + w_m x_m^i$$

$$\frac{\partial z}{\partial w_j} = x_j$$

Derivative of loss function

Score function:

$$z^i = w_0 x_0^i + \cdots + w_m x_m^i$$

$$\frac{\partial L}{\partial w_j} = (z - y)x_j$$

Derivative of loss function

Loss function:

$$L(\mathbf{x}^i; \mathbf{w}) = - (z^i \cdot y^i)$$

Compute $\frac{\partial L}{\partial w_j}$.

Gradient descent over a dataset

- Total loss function is the average loss over all examples:

$$L(\mathbf{x}; \mathbf{w}) = \frac{1}{n} \sum_i L^i(\mathbf{x}^i; \mathbf{w})$$

- What is $\nabla L_{\mathbf{w}}(\mathbf{x}; \mathbf{w})$?

- $$\nabla_{\mathbf{w}} L(\mathbf{x}; \mathbf{w}) = \frac{1}{n} \sum_i \nabla_{\mathbf{w}} L^i(\mathbf{x}^i; \mathbf{w})$$